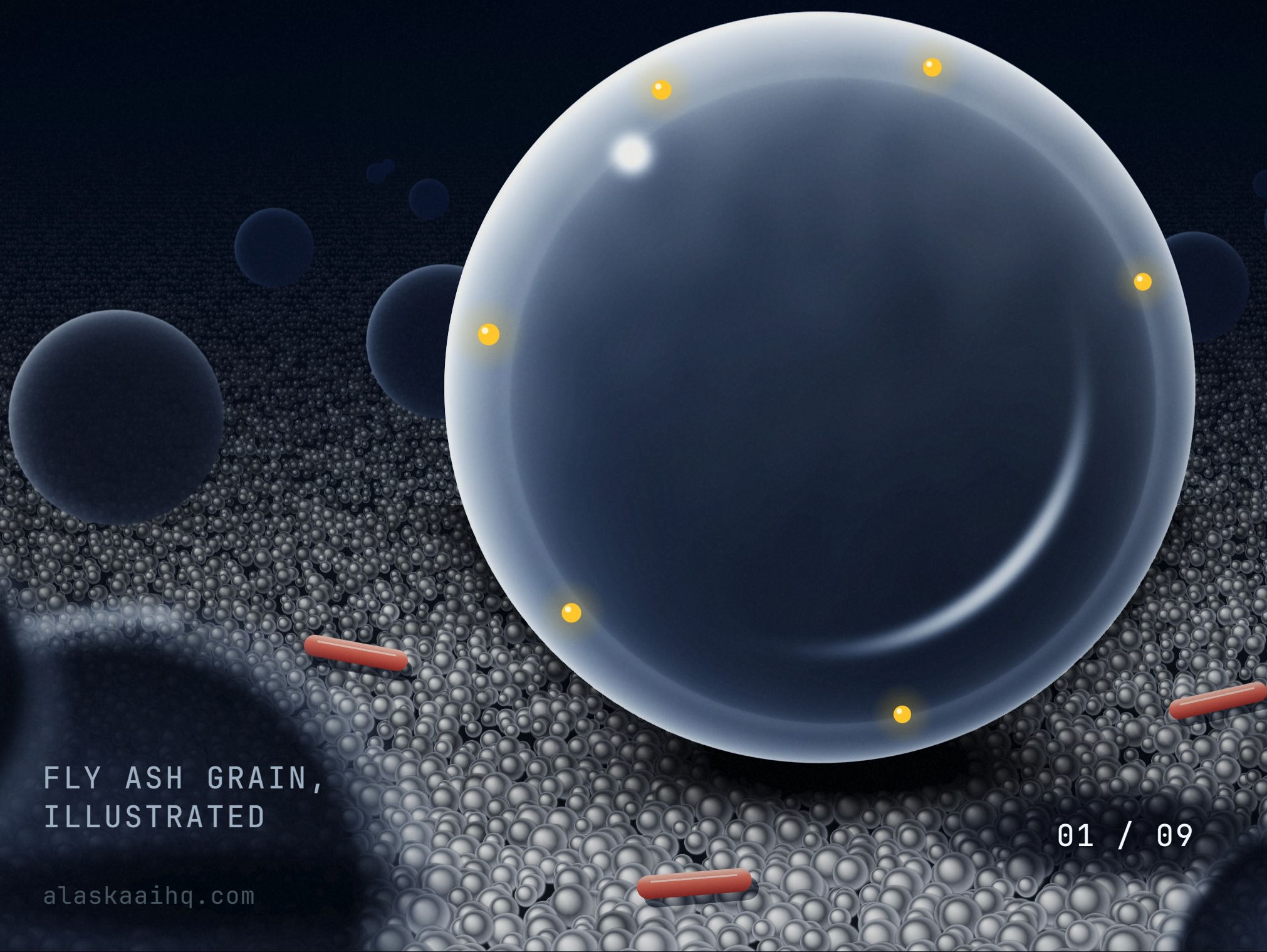


| | | | | | | | | |
GRAIN

ALASKA.AI

UAA · NSF · \$6 MILLION

Alaska's newest rare-earth AI won't hunt for ore. It will mind microbes.



FLY ASH GRAIN,
ILLUSTRATED

alaskaaihq.com

01 / 09

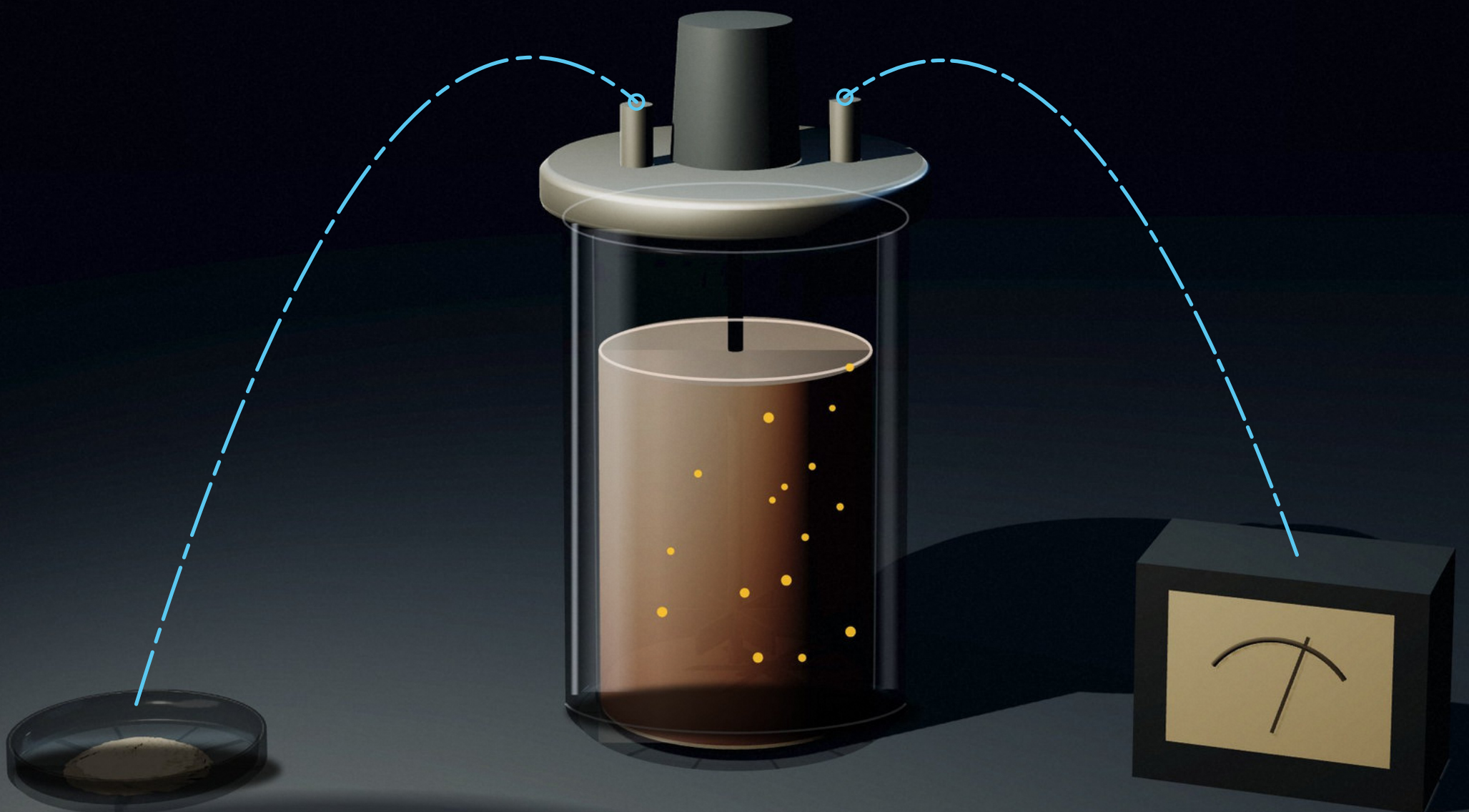
WHAT THE AWARD PLANS

Two jobs, and neither is prospecting.

A \$6 million NSF project led by UAA plans to pull rare earths out of coal refuse and ash with microbes. AI is to rank the waste first, then run the tank as power comes and goes.

JOB ONE • RANK THE FEED

JOB TWO • PACE THE POWER



BIOREACTOR, ILLUSTRATED

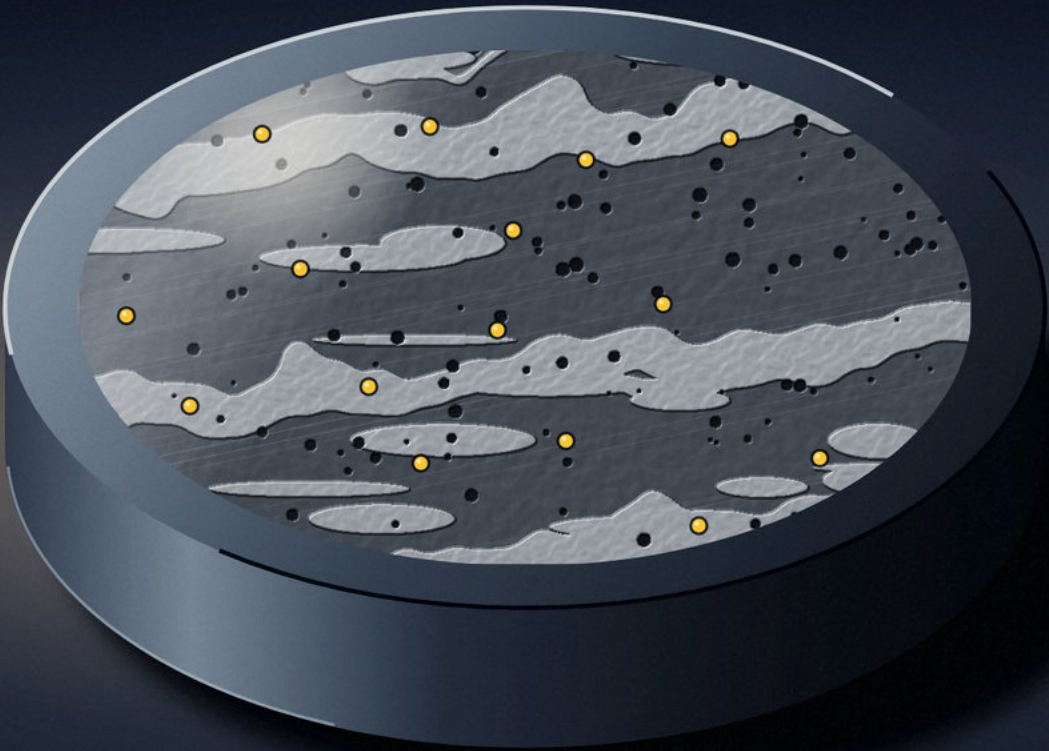
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EARLIER UAA LAB WORK

Alaska doesn't mine rare earths. Coal holds a little.



Researcher D'Lynn Gleason said so in earlier UAA lab work on coal donated by Usibelli Coal Mine in Healy. She said that coal carries rare earths, scandium and yttrium among them, too thin to mine with acid.



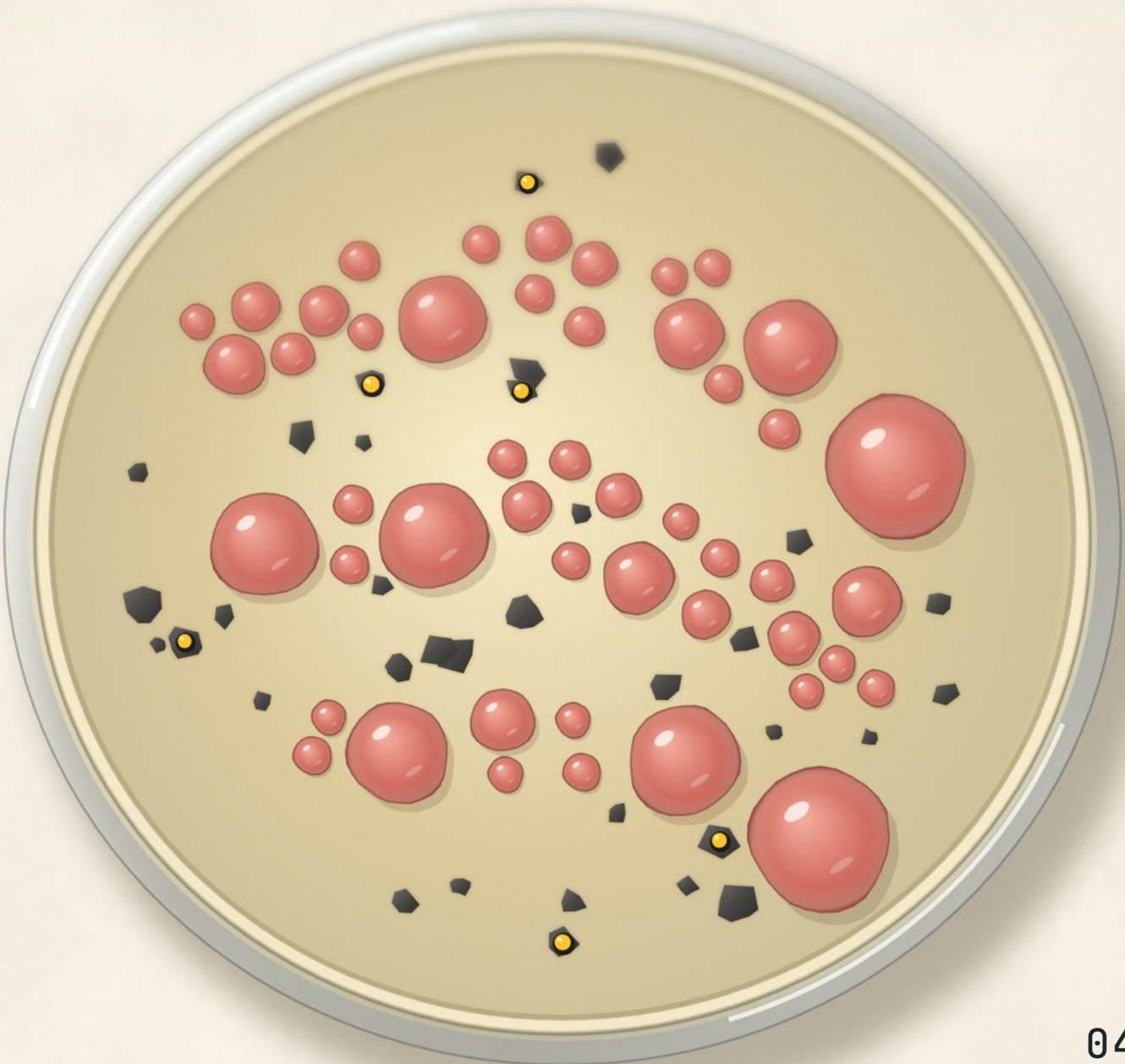
NOT THIS AWARD'S FEED

POLISHED COAL SECTION, ILLUSTRATED

One microbe is named.

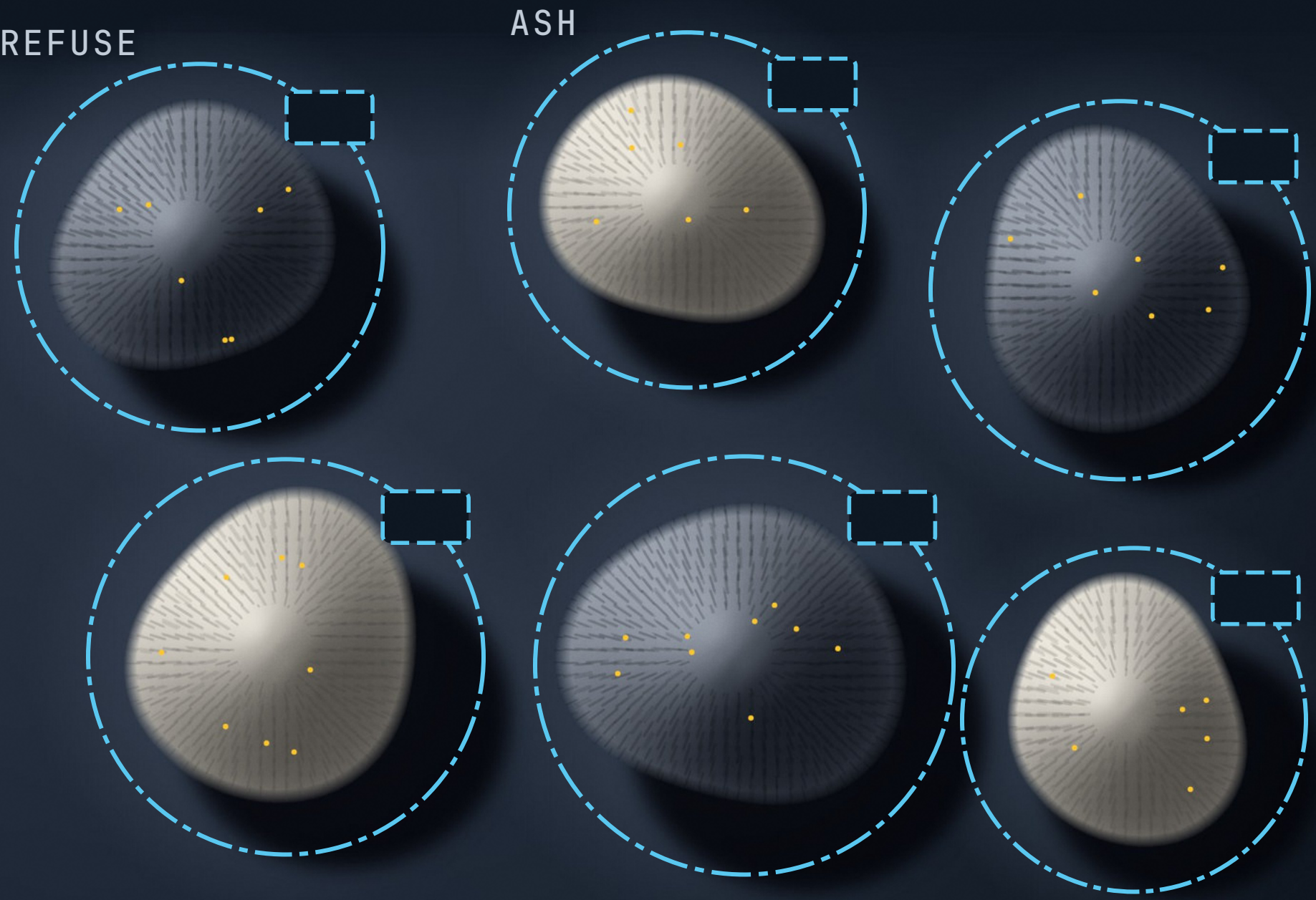
The award names *Shewanella oneidensis*. It will be tested on cost and life-cycle metrics.

SHEWANELLA ONEIDENSIS COLONIES, ILLUSTRATED



Rank the waste before it's fed.

UAF says AI will help predict how readily rare earths can be recovered from different materials. The award calls it a Virtual Assay to screen feedstocks. It is planned, not built.



DARK

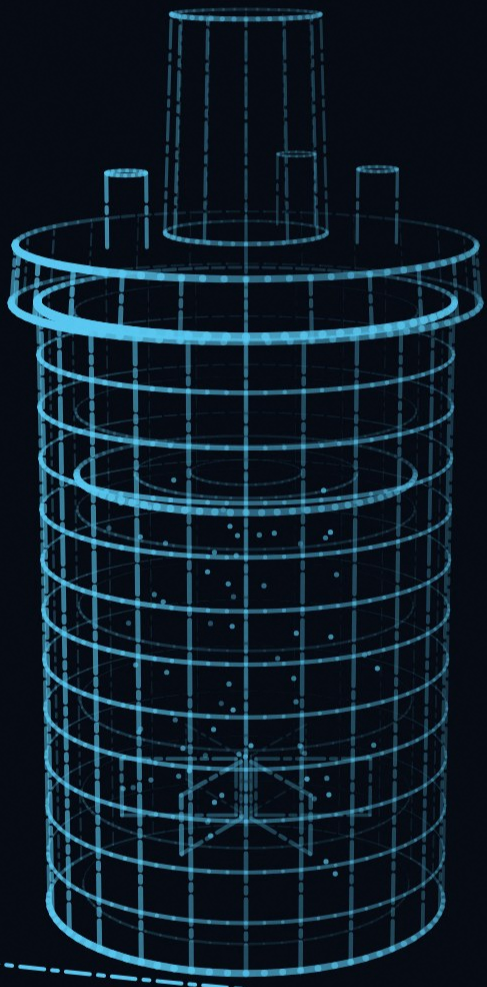
The power dips.
Something has to decide
what the tank does.



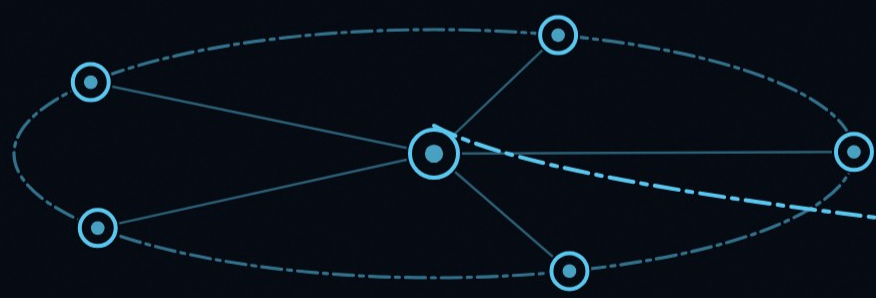
JOB TWO

Job two, the Alaska part.

Reinforcement-learning controllers will learn inside a digital twin of the bioreactor, linked to microgrid simulators, and adjust it as energy availability changes. The award's own word is energy-constrained.



DIGITAL TWIN • PLANNED



MICROGRID SIMULATOR



ASH SAMPLE

POWER AVAILABLE • SCHEMATIC



CONTROLLER • PLANNED

WHO HOLDS THE MONEY

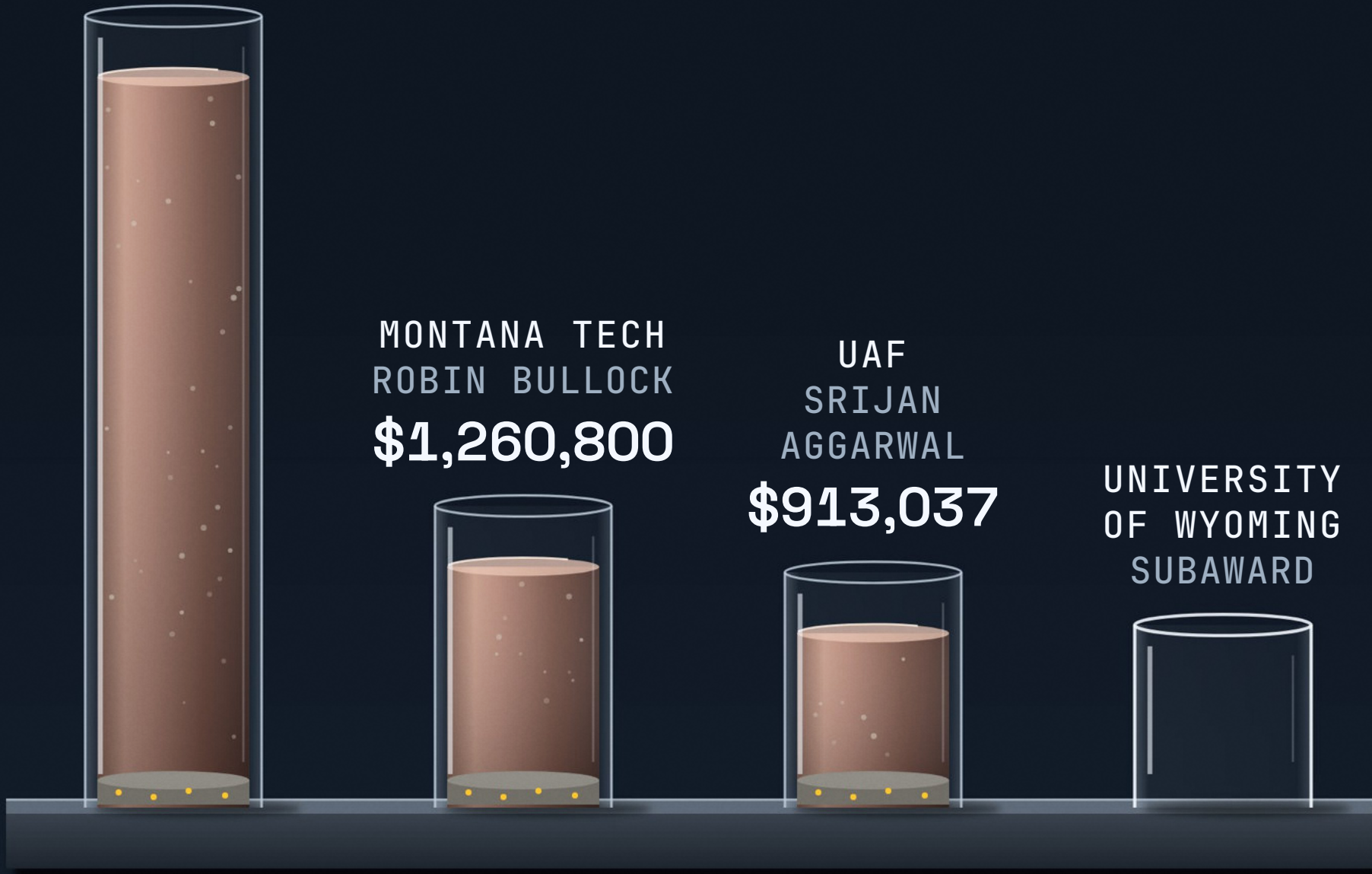
About \$6 million,
to August 31st, 2030.

UAA • LEAD
BRANDON BRIGGS
\$3,824,575

MONTANA TECH
ROBIN BULLOCK
\$1,260,800

UAF
SRIJAN
AGGARWAL
\$913,037

UNIVERSITY
OF WYOMING
SUBAWARD

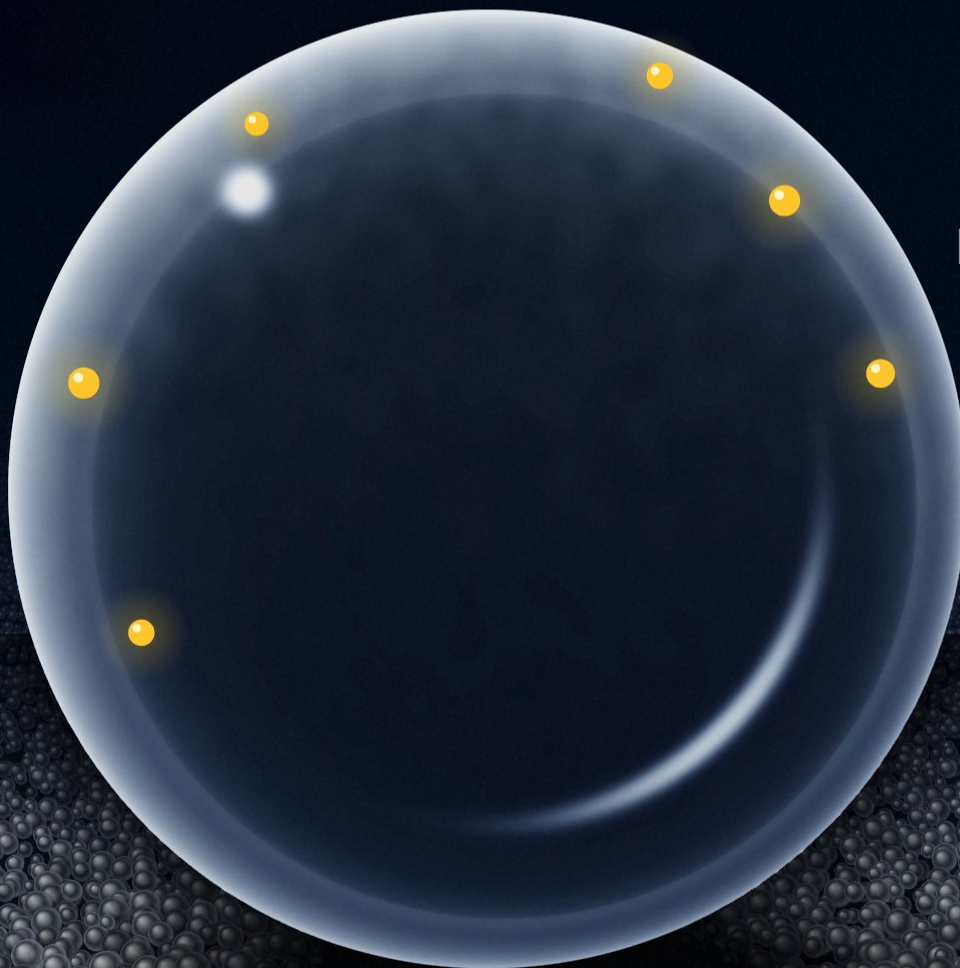


Began September 1st. One of eight EPSCoR projects sharing \$39.9 million.

NOT THIS AWARD

The UAF-led Critical Mineral Accelerator Engine gets \$15 million for its first two years and plans AI to locate deposits.

New deposits, or old waste? Where should Alaska point its mineral AI?



NOTHING RECOVERED YET

